

Name: _____

Date: _____

SOLUTIONS



Year 9 Mathematics

Diagnostic Test

CALCULATOR ALLOWED

1. Calculate:

a $21 + 19 + 10 + 30$ 80 b $168 - 47$ 121
c $1306 + 4683$ 5989 d $5867 - 2632$ 3235

2. At a concert, there are 1265 adults and 976 children. How many people in total are at the concert?

2241

3. Two friends each have a stamp collection. Jacqueline has 4568 stamps and Andrea has 7822 stamps. How many more stamps does Andrea have than Jacqueline?

3254

4. Calculate:

a $5 \times 4 \times 3 \times 2$ 120 b 15×24 360
c $4816 \div 4$ 1204 d $7028 \div 7$ 1004

5. In the AFL there are 16 football clubs. Each club selects 22 players for its senior team every week. How many senior players are picked each week across the whole AFL?

352

6. 7542 marbles are to be bagged. How many bags would there be if there were:

a 6 marbles in each bag? 1257b 18 marbles in each bag? 419

7. Calculate:

a 1754×13 22802 b $\frac{2769}{3}$ 923

8. Write each fraction in lowest terms.

a $\frac{8}{16}$ $\frac{1}{2}$

b $\frac{100}{300}$ $\frac{1}{3}$

9. Carry out these additions and subtractions. Give your answers in lowest terms.

a $\frac{3}{16} + \frac{5}{16}$ $\frac{8}{16} = \frac{1}{2}$

b $\frac{11}{16} - \frac{7}{16}$ $\frac{4}{16} = \frac{1}{4}$

c $\frac{3}{4} - \frac{2}{3}$ $\frac{1}{12}$

d $\frac{5}{6} + \frac{7}{12}$ $\frac{17}{12} = 1\frac{5}{12}$

e $2\frac{1}{4} + 1\frac{1}{2}$ $\frac{15}{4} = 3\frac{3}{4}$

f $2\frac{3}{4} - 1\frac{1}{8}$ $\frac{13}{8} = 1\frac{5}{8}$

10. Carry out these multiplications and divisions.

a $\frac{5}{7} \times \frac{3}{11}$ $\frac{15}{77}$

b $\frac{7}{8} \div \frac{3}{4}$ $\frac{7}{8} \times \frac{4}{3} = \frac{7}{6} = 1\frac{1}{6}$

c $3\frac{1}{2} \times 1\frac{1}{4}$ $\frac{7}{2} \times \frac{5}{4} = \frac{35}{8} = 4\frac{3}{8}$

d $4\frac{2}{3} \div 1\frac{1}{6}$ $\frac{14}{3} \div \frac{7}{6} = \frac{14}{3} \times \frac{6}{7} = 4$

e $\frac{5}{8} \times \frac{8}{11} \div \frac{3}{4}$ $\frac{5}{8} \times \frac{8}{11} \times \frac{4}{3} = \frac{20}{33}$

11. A hockey squad consists of 15 players. One-fifth of the squad is injured. How many players are fit to play?

$15 \times \frac{4}{5} = 12.$

12. Two-thirds of the students in a class of 27 are aged 13 years or older. How many students are less than 13 years of age?

$$27 \times \frac{1}{3} = 9$$

13. Convert these fractions to decimals.

a $\frac{4}{5}$

$$0.8$$

b $\frac{3}{7}$

$$0.428571$$

14. Calculate these sums and differences.

a $0.05 + 0.8$

$$0.85$$

b $7.6 + 0.3 + 0.1$

$$8$$

c $9.85 - 1.1$

$$8.75$$

15. Calculate these products and quotients.

a 0.3×0.4

$$0.12$$

b 0.03×0.4

$$0.012$$

c $24 \div 0.4$

$$60$$

d $2400 \div 0.04$

$$60000$$

16. What is the perimeter of a quadrilateral (4-sided figure) with side lengths 0.8 metres, 1.2 metres, 1.46 metres and 2.54 metres?

$$6 \text{ m.}$$

17. Round each number correct to 2 decimal places.

a 25.356

$$25.36$$

b 12.115

$$12.12$$

c 9.2346

$$9.23$$

d 1.005

$$1.01$$

18. Round these fractions and mixed numbers correct to 2 decimal places.

a $\frac{1}{3}$

0.33

b $5\frac{3}{8}$

3.38

c $7\frac{1}{7}$

7.14

19. Arrange these numbers from smallest to largest.

0.002, 2.02, 22.202, 2.222, 22.0002

0.002, 2.02, 2.222, 22.0002, 22.202.

20. Convert these decimals to fractions or mixed numbers.

a 0.6

$\frac{3}{5}$

b 0.125

$\frac{1}{8}$

c 2.375

$2\frac{3}{8}$

21. Calculate:

a $-3 + 4$

1

b $-7 + 3$

-4

c $-5 + (-8)$

-13

d $-13 + 10$

-3

22. Calculate:

a $-5 - 2$

-7

b $2 - 5$

-3

c $-3 - (-8)$

5

d $7 - (-7)$

14

23. The temperature in a mountain township went from 8°C to -2°C . What was the change in temperature?

10°C

24. Write the answers to these multiplications and divisions.

a $6 \times (-3)$

-18

b $-3 \times (-2)$

6

c $-12 \div (-2)$

6

d $-8 \div (-4)$

2

25. Evaluate:

a	$(-5)^2$	<u>25</u>	b	$-(-2)^2$	<u>-4</u>
c	$-(3-5)$	<u>2</u>	d	$-3 \times (-5+8)$	<u>-9</u>
e	$-3-5 \times 20$	<u>-103</u>			

26. a Evaluate $5a$ when $a = 19$. 95

b Evaluate $-17m$ when $m = 4$. -68

c Evaluate $6c + 3d$ when $c = 5$ and $d = -2$. 24

27. Simplify each expression by collecting like terms.

a $2x + 5y - 3y + 4x$ $6x + 2y$

b $3a + 4b^2 - a + 2b$ $2a + 2b + 4b^2$

c $15m - 4n + 9mn - 6m + 3n$ $9m - n + 9mn$

d $xy^2 - 4 + 3x^2 + 2xy^2 + 8$ $3xy^2 + 3x^2 + 4$

28. Simplify each expression.

a	$2x \times 5$	<u>$10x$</u>	b	$6a \times -3b$	<u>$-18ab$</u>
c	$2xy \times 4xz$	<u>$8x^2yz$</u>	d	$12y \div 3$	<u>$4y$</u>
e	$48g^2 \div 6$	<u>$8g^2$</u>	f	$4xy \div 2x$	<u>$2y$</u>
g	$\frac{9mn}{3m}$	<u>$3n$</u>	h	$\frac{2a}{5} \times \frac{7c}{4}$	<u>$\frac{7ac}{10}$</u>

29. Expand the brackets.

a	$2(5x+7)$	<u>$10x+14$</u>	b	$y(7-3x)$	<u>$7y-3xy$</u>
c	$-5(4x+9)$	<u>$-20x-45$</u>	d	$-3a(3a-2)$	<u>$-9a^2+6a$</u>

30. Expand the brackets and collect like terms.

a	$5(x-2)+7$	<u>$5x-3$</u>	b	$3(4x+1)-5x$	<u>$7x+3$</u>
c	$4x(1+5x)-2$	<u>$4x+20x^2-2$</u>	d	$2(2x+3)+3(x+4)$	<u>$7x+18$</u>
		$4x+20x^2-2$			$4x+6+3x+12$

31. Expand the brackets and collect like terms.

a $(x+1)(x+2)$ x^2+3x+2 b $(x-2)(x+3)$ x^2+x-6
 c $(3x+5)(2x+1)$ $6x^2+13x+5$ d $(x+2y)(3x+y)$ $3x^2+7xy+2y^2$

32. Calculate these squares without using your calculator.

a 6^2 36 b 8^2 64
 c 13^2 169 d 20^2 400

33. Calculate the positive value of these square roots without using your calculator.

a $\sqrt{4}$ 2 b $\sqrt{81}$ 9
 c $\sqrt{196}$ 14 d $\sqrt{625}$ 25

34. Round each of these numbers to 3 decimal places.

a 2.32419 2.324 b 14.0278212 14.028

35. Convert each percentage to a fraction in simplest form.

a 66% $\frac{66}{100} = \frac{33}{50}$ b 35% $\frac{35}{100} = \frac{7}{20}$
 c 3.2% $\frac{32}{1000} = \frac{4}{125}$ d 448% $\frac{448}{100} = 4\frac{12}{25}$

36. Express each of the following as a percentage.

a 0.7 70% b 5.68 568%
 c $\frac{7}{25}$ 28% d $\frac{9}{4}$ 225%

37. Calculate:

a 14% of 120 16.8 b 73% of 5200 3796

38. a Express \$1.20 as a percentage of \$6.00 $1.2 \div 6 \times 100 = 20\%$
 b Express 630 metres as a percentage of 4.5 kilometres. $630 \div 4500 \times 100 = 14\%$

39. The kangaroo population at a National Park is surveyed and found to be 15 000 kangaroos.

Three months later, the kangaroo population had increased to 17 400. Find the percentage increase in the population.

$17400 - 15000 = 2400$ $\therefore \frac{2400}{15000} \times 100 = 16\%$

40. A DVD player has a marked price of \$240. If it is discounted by 12%, what is the sale price?

$$240 \times 88 \div 100 = \$211.20$$

41. Factorise each of the following expressions by first finding a common factor.

a $3x + 12$ $3(x + 4)$

b $4a - 28$ $4(a - 7)$

c $x + 5x^2$ $x(1 + 5x)$

d $8 - 40y$ $8(1 - 5y)$

e $25z^2 + 15z$ $5z(5z + 3)$

f $18y^3 - 3y$ $3y(6y^2 - 1)$

g $-32p^2q + 48pq^2$ $8pq(-4p + 6q)$

h $-56q^2 - 28pq - 14p^2q$ $14q(-4q - 2p - p^2)$

42. Phuong is 5 cm taller than Marina. If Phuong is x cm tall, write an expression for Marina's height.

$$x - 5$$

43. Circle the mathematical expression that corresponds to the following statement:

'2 more than 5 times x '

i $x + 10$

ii $5x + 2$

iii $5(x + 2)$

iv $2x - 5$

v $2x + 5$

44. Solve:

a $a + 3 = 7$ $a = 4$

b $4d = 12$ $d = 3$

c $q - 7 = 15$ $q = 22$

d $\frac{t}{5} = 6$ $t = 30$

e $3x - 15 = 12$ $x = 9$

f $\frac{7x}{5} = 14$ $x = 10$

g $6p - 4 = -22$ $p = -3$

h $4x - 6 = x + 12$ $x = 6$
 $3x = 18$

45. Solve:

a $5(x + 12) = 75$ $x = 3$
 $x + 12 = 15$

b $5(4x - 2) = 50$ $x = 3$
 $4x - 2 = 10$

c $4y + \frac{1}{2} = \frac{3}{2}$ $y = \frac{1}{4}$

d $\frac{3}{5}(n + 2) = \frac{6}{5}$ $n = 0$
 $n + 2 = 2$

e $m + 3.2 = 5.9$ $m = 2.7$

f $1.2(s + 4) = 8.4$ $s = 3$
 $s + 4 = 7$

46. A number is multiplied by 4 and 8 is subtracted from the result; the final result is 16. What is the original number?

$$4x - 8 = 16$$

$$4x = 24$$

$$x = 6$$

47. a What is the base of the power 6^5 ? 6

b What is the index of the power 8^3 ? 3

48. Simplify the following. Leave your answer as a single power.

a $2^6 \times 2^5$ 2^{11} b $5^8 \times 5^4$ 5^{12} c $x^1 \times x^4$ x^5

49. Simplify the following. Leave your answer as a single power.

a $3^{14} \div 3^2$ 3^{12} b $\frac{11^6}{11^3}$ 11^3 c $y^8 \div y^3$ y^5

50. Write each expression as a single power.

a $(2^3)^2$ 2^6 b $(5^4)^3$ 5^{12} c $(a^5)^2$ a^{10}

51. Simplify:

a $a^2b^3 \times a^4b$ a^6b^4 b $3x^5 \times 2x^3$ $6x^8$ c $(2m^3n^2)^3$ $8m^9n^6$
d $\frac{a^5b^3}{a^2b^2}$ a^3b e $6m^7n^4 \div 2m^3n^2$ $3m^4n^2$ f $\left(\frac{x^5}{x^2}\right)^4$ $= (x^3)^4 = x^{12}$

52. a If 3 kg of apples cost \$7.50, find the cost of 7 kg of apples. $7.5 \div 3 \times 7 = \$17.50$

b If 3 metres of fabric cost \$48.00, find the cost of 5 metres. $48 \div 3 \times 5 = \$80$

c If 6 maths books cost \$84.00, find the cost of 9 maths books. $84 \div 6 \times 9 = \$126$

d If 4 bags of chips cost \$6.60, find the cost of 10 bags of chips. $6.60 \div 4 \times 10 = \16.50

53. A cyclist travelled at 27 km/hr for 2 hours. How far did she go? 54 km.

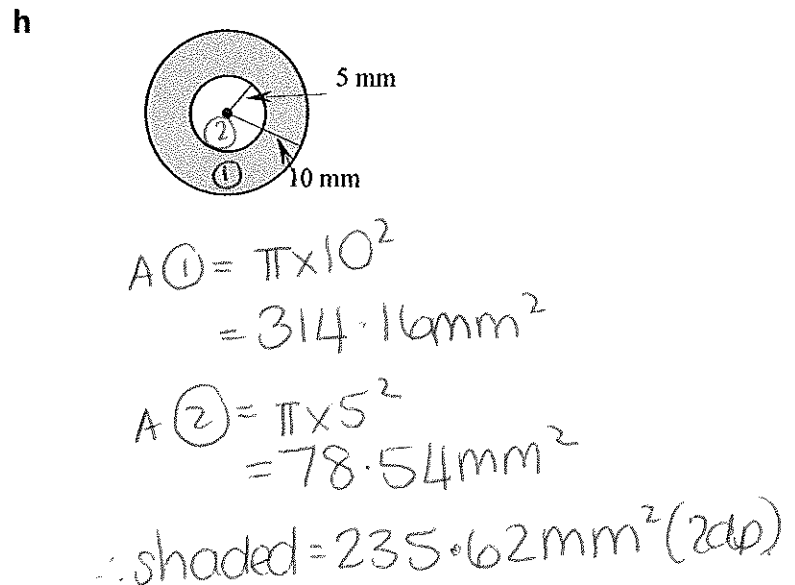
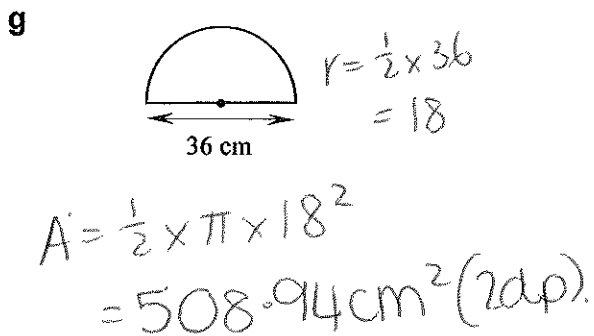
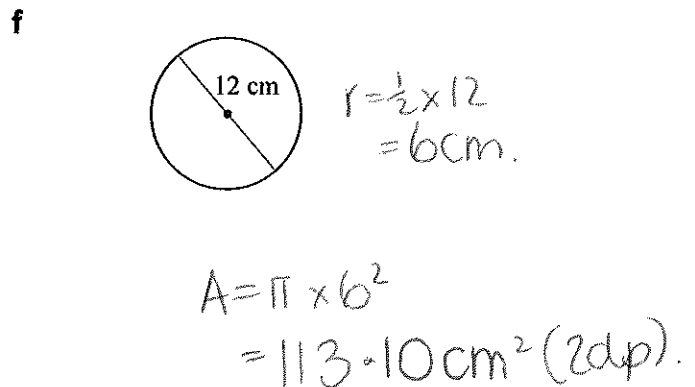
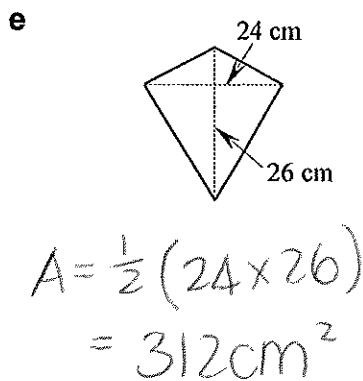
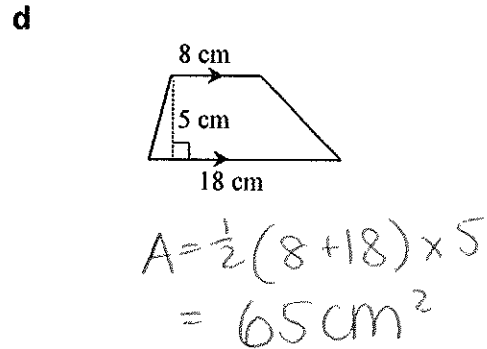
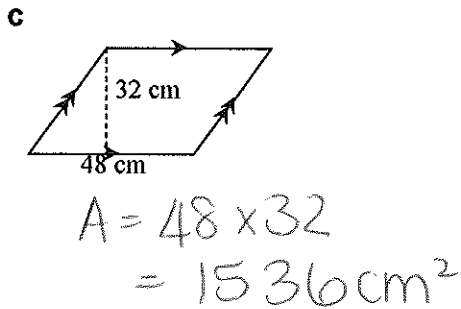
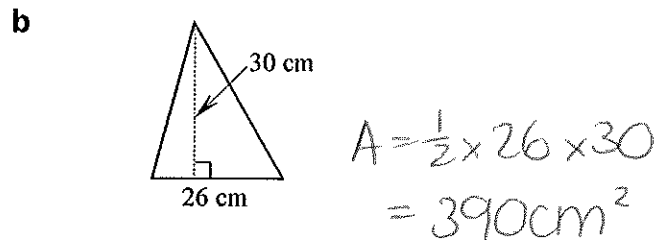
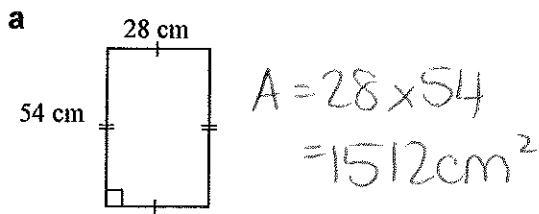
54. If a bus travels a distance of 50 km at an average speed of 80 km/hr, how long does the trip take?

$80 \text{ km} = 60 \text{ min}$ $x = 50 \div 80 \times 60 = 37\frac{1}{2} \text{ min.}$
 $50 \text{ km} = x.$

55. A truck travelling at a constant speed travels 285 km in 3 hours. Assuming it maintains its constant speed, how far will it have travelled in 4 hours?

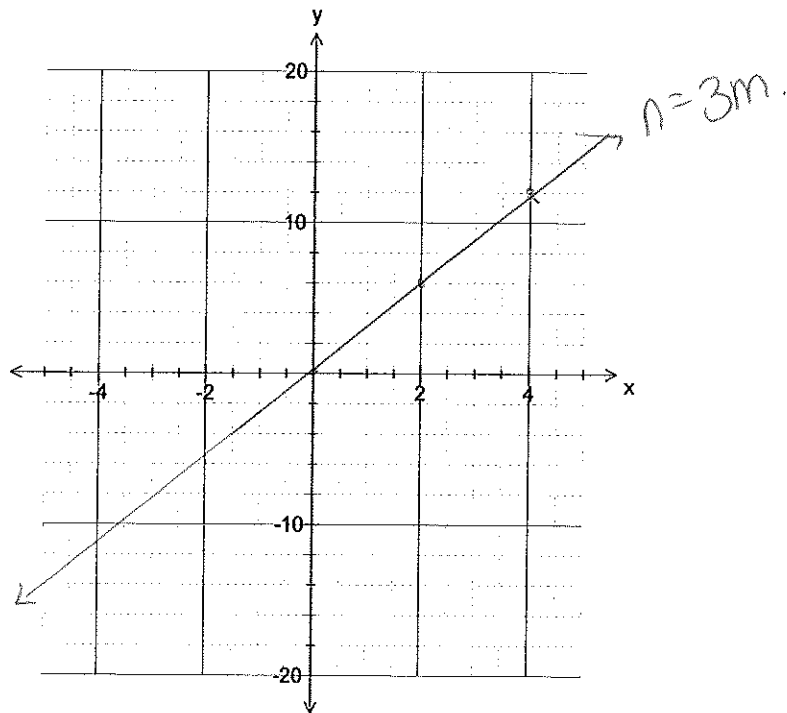
$285 \div 3 \times 4 = 380 \text{ km.}$

56. Calculate the area of the following figures.



57. Given the formula $n = 3m$, complete the table and plot the corresponding points on a Cartesian plane, connecting them with a straight line.

m	0	1	2	3	4
n	0	3	6	9	12



58. Given the formula $q = \frac{1}{2}p$, complete the table and plot the corresponding points on a Cartesian plane, connecting them with a straight line.

p	-2	-1	0	1	2
q	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1

